

SRISHTI DIXIT

+917985742575 ♦ Varanasi, India

srishtidixit.rs.phy18@itbhu.ac.in ♦ [Linkedin](#) ♦ [Google Scholar](#)

PROFESSIONAL SUMMARY

An accomplished and experienced researcher specializing in experimental condensed matter physics, with research experience in magnetic and quantum materials. My work includes conducting and analyzing experiments. I have good communication and writing skills that help me clearly explain scientific results. I have practical experience with instruments such as MPMS, PLD, XRD, and low-temperature cryostats. I have expertise in analysing the transport, magnetic properties and different techniques for crystal growth of single crystals, bulk systems and thin films. I have basic knowledge of Python, High Performance Computer (HPC) clusters and basic knowledge of Density Functional Theory (DFT).

EDUCATION

Doctor of Philosophy (Ph.D.), Dept. of Physics, Indian Institute of Technology (BHU) Varanasi, India 2018-2023

- **Viva-voce:** 15 May 2024 (Degree Awarded)
- **Thesis title:** Study of Magnetic and magneto-transport properties of some Heusler alloys. **(9.13/10)**
- Relevant Coursework: Advanced Condensed Matter Physics and Materials Science, Advanced Techniques for Materials Characterisation, Advanced Experimental Physics, Low Dimensional Physics.

Master of Science, Chhatrapati Shahu Ji Maharaj University, Kanpur, India July 2014- June 2016
Major in Physics Specialization in Electronics.

Bachelor of Science, Chhatrapati Shahu Ji Maharaj University, Kanpur, India 2011- May 2014
Relevant Subjects: Physics, Chemistry, Maths

Higher Secondary (10+2), Shivaji Inter College, Kanpur, India July 2009- June 2011
The Board of High School and Intermediate Education Uttar Pradesh, Allahabad (India)

Secondary School (10), Dr. Virendra Swaroop Education Centre, Kanpur, India July 2009
The Board of High School and Intermediate Education Uttar Pradesh, Allahabad (India)

SKILLS

Instrumentation	X-ray Diffraction (XRD), Magnetic Property Measurement System (MPMS) Physical properties measurements system, Pulse Laser Deposition (PLD) Vacuum Pump/ Quartz Tube Sealing, Nanovoltmeter, SourceMeter, AC and DC current Source
Soft Skills	Density Functional Theory (DFT), Python
Software	Quantum Espresso, HighScore Plus (XRD Analysis), LABVIEW
Graph Analyzer and Plotting	Gnu plot, Origin, Xmgrace, Origin software

WORK EXPERIENCE

Research Scholar Jul 2018 - 29 Dec 2023
Indian Institute of Technology (BHU) *Varanasi, India*

- Conducting extensive research in synthesizing and Fabrication of various bulk systems using **Arc melting method** and **Solid-state reaction** methods.
- Employed Modified Bridgmann Technique for single crystals' growth.
- Fabrication of Thin Film using Pulse Laser Deposition (**PLD**) under varying condition like thickness, pressure and temperature.

- Demonstrated the structural phase from the XRD patterns of bulk systems, single crystals and thin films also.
- Examined some fascinating magnetic properties like memory effect (in hold and run way), **Chi AC**, **MT** and **MH** in Heusler alloys(HAs).
- applying spectroscopy techniques like **Raman spectroscopy**, **XAS/XMCD**, **XPS**, and **SEM**, for detailed analysis.
- Evaluated the **Seebeck coefficient**, Power Factor (PF), Thermal Conductivity and Figure of merit (ZT) in Topological Insulator (TI) and HAs.
- Discovered the Spin- Valve effect in HAs after analysing the magnetoresistance (MR)in HAs.
- Discovered Anomalous Hall effect in HAs after examining the transport properties.
- Performed DFT of magnetic system including convergence test, Band structure and density of states calculation.

HONORS AND AWARDS

National Eligibility Test (NET) Rank-4 (All India Rank), Awarding institution: CSIR-UGC-NET, New Delhi, India

Graduate Aptitude Test in Engineering (GATE) Rank- 711 (All India Rank)

LANGUAGES

Hindi (Native)

English (Full Professional Proficiency)

RESEARCH PUBLICATION

Published

- **Existence of exotic magnetic phases along with exchange bias and memory effect in frustrated beta-Mn Heusler alloy.**
Srishti Dixit, Labanya Ghosh, Mohd Alam, Satya Vijay Kumar, Neha Patel, Swayangsiddha Ghosh, Nisha Shahi , and Sandip Chatterjee
 Journal of Applied Physics, (2023), Vol.: 133, Page: 013904, [DOI](#)
- **Raman effect and unusual transport properties of narrow band gap semiconductor Heusler alloy.**
Srishti Dixit, Swayangsiddha Ghosh, Neha Patel, Mohd Alam, Krishanu Bandyopadhyay, Nisha Shahi, Yogendra Kumar, M. Sawada, K. Shimada, Satyen Saha, Sanjay Singh and Sandip Chatterjee
 Europhysics Letter (2023), Vol.: 144, Page: 56003, Issue: 5, [DOI](#)
- **Effect of antisite disorder on magnetic and transport properties of quaternary Heusler alloys**
Srishti Dixit,Swayangsiddha Ghosh, Sanskar Mishra, Nisha Shahi Prashant Shahi, Sanjay Singh, A.K. Bera, S.M. Yusuf, Yoshiya Uwatoko, C-F Chang and Sandip Chatterjee.
 Arxiv (2024), [DOI](#)
- **Anharmonic phonon interactions and Kondo effect in $FeSe/Sb_2Te_3/FeSe$ hetero-structure: Proximity effect between ferromagnetic chalcogenide and di-chalcogenide.**
 Labanya Ghosh, Mohd Alam, Mahima Singh,**Srishti Dixit**, Satya Vijay Kumar, Abhineet Verma, Prashant Shahi
 Nanoscale (2022),Vol.: 14, Page: 10889-10902, Issue: 30, [DOI](#)
- **Extremely large magnetoresistance with coexistence of a nontrivial Berry phase in $Nb_{0.5}Ta_{0.5}P$: an experimental and theoretical study.**
 Vinod K Gangwar, Saurabh Singh, Swayangsiddha Ghosh, Srishti Dixit, Shiv Kumar, Prashant Shahi, Yoshiya Uwatoko, Sandip Chatterjee
 Journal of Materials Chemistry C (2024), Vol: 12, Page: 16375-16388, Issue: 40, [DOI](#)

- **Coexistence of Kondo effect and weak anti-localisation in Topological Insulator/Ferromagnetic heterostructure.**
Labanya Ghosh, Mohd Alam, **Srishti Dixit**, Swayangsiddha Ghosh, Archana Tiwari, Prashant Shahi, Yoshiya Uwatoko and Sandip Chatterjee
Applied Surface Science (2024), Vol.: 673, Page: 160850, [DOI](#).
- **Double glassy states and large spontaneous and conventional exchange bias in $La_{1.5}Ca_{0.5}CoFeO_6$ ferrimagnetic double perovskite.**
Prajyoti Singh, Rahul K. Singh, **Srishti Dixit**
Journal of Physics:Condensed Matter (2022), Vol.: 34, Page: 375803, Issue 37, [DOI](#)
- **Observation of the multiple magnetic phases in double perovskite $Pr_{1.8}La_{0.2}CoFeO_6$.**
Dheeraj Kumar, Prajyoti Singh, Mohd Alam, Satya Vijay Kumar, **Srishti Dixit**, Krishanu Bandyopadhyay, GD Dwivedi, Satyen Saha, AK Ghosh, Sandip Chatterjee
Journal of Magnetism and Magnetic Materials (2024), Vol.: 608, Page: 172408, [DOI](#)
- **Electrical and magnetic properties of double perovskite $Y_{2-x}Ca_xCoMnO_6$ ($x = 0.1, 0.2, 0.5$).**
Dheeraj Kumar, Mohd Alam, Prajyoti Singh, Srishti Dixit, A.K. Ghosh, Sandip Chatterjee
Journal of Materials Science: Materials in Electronics (2024), Vol. 35, Page 1670, Issue 25, [DOI](#)
- **Lattice dynamics of $Bi_{1.9}Dy_{0.1}Te_3$ topological insulator.**
Labanya Ghosh, Vinod K. Gangwar, Mahima Singh, Satya Vijay Kumar, **Srishti Dixit**, Abhineet Verma, Durgesh Kumar Sharma
Physica B: Condensed Matter (2022), Vol: 640, Page: 414050, [DOI](#)
- **Multifunctional behaviour in B-site disordered double perovskite $EuPrCoMnO_6$.**
Mohd Alam, Labanya Ghosh, Supriyo Majumder, Prajyoti Singh, Satya Vijay Kumar, **Srishti Dixit**, Dheeraj Kumar, Khyati Anand, Seema Kumari, A K ghosh, R J Choudhary and Sandip Chatterjee
Journal of Physics D: Applied Physics (2021), Vol.: 55, Page: 255003, Issue: 25, [DOI](#)
- **Enhancement of multiferroic and optical properties in $BiFeO_3$ due to different exchange interactions between transition and rare earth ions.**
Seema Kumari, Khyati Anand, Mohd Alam, Labanya Ghosh, **Srishti Dixit**, Rahul Singh, Anil Kumar Jain, Seikh Mohammad Yusuf, Chetna Gautam, Anup K Ghosh, Anita Mohan, Sandip Chatterjee
physica status solidi (b) (2023), Vol.: 260, Page: 2300026, issue: 9 [DOI](#)
- **Giant dielectric constant, magnetocaloric effect and spin-phonon coupling in $EuTbCoMnO_6$ semiconductor.**
Mohd Alam, Labanya Ghosh, **Srishti Dixit**, Madhusmita Jena, Seema Kumari, Satya Vijay Kumar, Dheeraj Kumar, Abhineet Verma, A. K. Ghosh, Satyen Saha, R. J. Choudhary, Sandip Chatterjee
Physica B: Condensed Matter (2023), Vol.: 665, Page: 415043, [DOI](#)
- **Spin-phonon coupling and giant dielectric constant in $Bi_{0.5}La_{0.5}Fe_{0.4}Al_{0.1}Mn_{0.5}O_3$.**
Seema Kumari, Mohd Alam, **Srishti Dixit**, Labanya Ghosh, Swayangsiddha Ghosh, Prince K Gupta, Jayraj Rohit, Khyati Anand, Nabakumar Rana, Suchandra Mukherjee, Anita Mohan, Aritra Banerjee and Sandip Chatterjee
Journal of Materials Science: Material in Electronics(2023), Vol.: 34 , Page: 2142, [DOI](#)

Under review and Under preparation

- **Coexistence of spin valve effect and anomalous Hall effect in $Ni_2MnAl_{0.5}Sn_{0.5}$**
Srishti Dixit, Swayangsiddha Ghosh, Labanya Ghosh Sanskar Mishra, Nisha Shahi Prashant Shahi, Sanjay Singh Archana Tiwari and Sandip Chatterjee.
- **Chemical Pressure Enhanced Low-Field Modulated Exotic Magnetic Topological Phase**
Labanya Ghosh, **Srishti Dixit**, Mohd Alam, Neha Patel, Swayangsiddha Ghosh and Sandip Chatterjee
Communicated in Physical review B

CONFERENCES AND WORKSHOPS

1. Faculty Development programme on Materials Characterization

Hybrid Mode, India, 2022

2. 65th DAE Solid State Physics Symposium (Participation Certificate)

Bhabha Atomic Research Centre (BARC), Mumbai, India, 2021

3. 7th Conference on Neutron Scattering (Participation Certificate)

Hybrid mode, Mumbai, India, 2021

4. International Webinar on material synthesis and Characterization

Dr. Ram Manohar Lohia Awadh University U.P, India and Department of Natural Sciences Florida Polytechnic University, U.S.A, 2020,

5. 1st International e-Conference on Recent Advances in Physics & Materials Science

Kurseong College, Darjeeling, West Bengal, In collaboration with St. Joseph's College, Darjeeling, West Bengal, India, 2020

6. IX International Conference on perspectives in Vibration Spectroscopy

UGC-DAE Consortium for Scientific Research, Indore & Devi Ahilya Vishwavidyalaya, Indore, India, Dec, 2020

7. 33rd AGM of MRSI and 4th Indian Material Conclave IUMRS-ICA 2022

Indian Institute of Technology, Jodhpur, Rajasthan. 2022

LEADERSHIP

I have had the privilege of guiding and mentoring numerous students, including new Research Scholars., M.Sc., M.Tech., and B.Tech. candidates, in their projects and final year thesis. With a focused vision and strategic approach, I empower my research group to tackle challenges head-on and devise inventive solutions. I am dedicated to nurturing emerging talent, fostering innovation, and delivering exceptional outcomes through proactive problem-solving and effective project management.